Evaluation, Deployment and Monitoring of Models

Training a model is the easy part. Deciding whether it is good enough — for whom, for what and for how long — is the hard part, and the subject of this book.

It follows a model through the rest of its life: how to evaluate it honestly, with sound protocols, the right metrics and statistical comparisons; how to combine models with each other and with expert knowledge, and when a model should not predict at all; how to make models fair, explainable, secure and compliant with the European AI Act; how to automate data science, with and without language models; how to deploy models with APIs, pipelines and tests; and how to monitor them when the world changes, with drift detection, A/B tests and bandits.

Every unit includes Python code that was run to produce the results shown, and exercises with solutions.

1 Model evaluation 2 Hybrid models 3 Reliable models 4 Learning to learn
5 Deployment 6 Monitoring and updating 7 Applications

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